

Mykhailo Kapustin

AI/ML Engineer

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Summary

Full-stack engineer and researcher with 10+ years across frontend, backend, and DevOps, with the last 2–3 years focused on production LLM/multi-agent systems and graph-based knowledge representation. I lead engineering at ASRP, an independent AI research lab I run on an ongoing basis — its R&D on agent architectures and knowledge graphs directly feeds the quality of what I deliver on client work. EU citizen, Berlin-based, open to full-time, contract, or freelance engagements.

Skills

AI / ML	LangChain, LangGraph (3 yrs, production), knowledge graphs, RAG, multi-agent systems, Google ADK, Vertex AI, Gemini, OpenAI/GPT-4, PyTorch
Backend	Node.js, Python/FastAPI, GraphQL, tRPC, WebSockets
Frontend	Vue.js, Nuxt.js, Pinia, React
Cloud / DevOps	Docker, Terraform, CI/CD, AWS, GCP, Azure
Databases	PostgreSQL, Redis, MongoDB, Pinecone

Experience

Lead Engineer — ASRP (independent AI research lab) Apr 2023 – Present

- Built a transatlantic research holding from scratch: 10+ products across EdTech, AI, media, and blockchain
- Built the first prototype of a LangChain-based forecasting platform (GFS) at an AI hackathon in Berlin (Theta EuroCon)

Selected client engagements (delivered through ASRP)

Multi-tenant AI-organizations platform — a German AI SaaS company

- NestJS + React + PostgreSQL with row-level security (FORCE RLS), JWT tenant context, and invite-only auth; one configured agent deploys across three channels — website widget, Outlook add-in, and email processor
- Natural-language agent authoring: describe an agent in plain text, get a working, deployed agent back in ~1 minute
- Top contributor (~45%) across both core repositories; owned the staging → production release flow

Voice-AI and telephony platform — a US logistics-tech company

- Built a conversational voice agent for inbound carrier calls that is non-hallucinating by construction — the backend owns all data and decisions, the agent only converses and reports; decomposed a monolithic agent into a multi-agent system
- Built a cloud-PBX integration with an asymmetric source-of-truth model (PBX authoritative for live state, DB for UI reads), SIP-Call-ID call correlation, and a Redis buffer-and-replay fix for a fast-call event race
- Built a ring-all call router using Redis Redlock for deterministic first-to-answer claims under concurrent-answer races (TypeScript, Node.js, Redis, WebSocket, Jambonz)

Earlier roles (2016–2023): Team Lead / Senior Backend Developer (MOB.325 / LAB325, Ukraine — Node.js, GraphQL, Docker, AWS, ClickHouse), Team Lead (Coelix, Tel Aviv, remote), Senior Backend Developer (Provectus, San Francisco), Full Stack Developer (Ephyros, HYS Enterprise, Top Agent, PHP-academy), Backend Developer (SuperMediaAds) — PHP / Laravel / Vue foundation years.

Publications

11 technical articles on AI agents, infrastructure, and NLP — ~124K combined reads across Habr, Medium, and Hashnode. Selected:

- [Docling in Working with Texts, Languages, and Knowledge](#) — 13.9k views. Documents as graphs of typed, linked nodes; LLM-driven graph querying.
- [Building AI Agent Infrastructure](#) — 10k+ views. Filesystem Registry → Package Manager → AgentOps/MCP&A2A — scaling multi-agent systems.
- [My Personal Exam: MVP LLM Agent on Google ADK](#) — 7.8k views. Modular multi-agent architecture, reviewer-agent validation patterns.
- [Google ADK and "Startup Technical Guide: AI Agents"](#) — 5.3k views. Grounding, GraphRAG, and AgentOps for production-reliable agents.

Education

Master's, Automation and Computer-Integrated Technologies — State University of Intelligent Technologies and Telecommunications (2017–2022)

Work Authorization

EU citizenship — no visa/work-permit sponsorship required in the EU. Berlin-based. Open to full-time, contract, or freelance engagements — also able to contract directly through US- and Kazakhstan-based entities where useful.